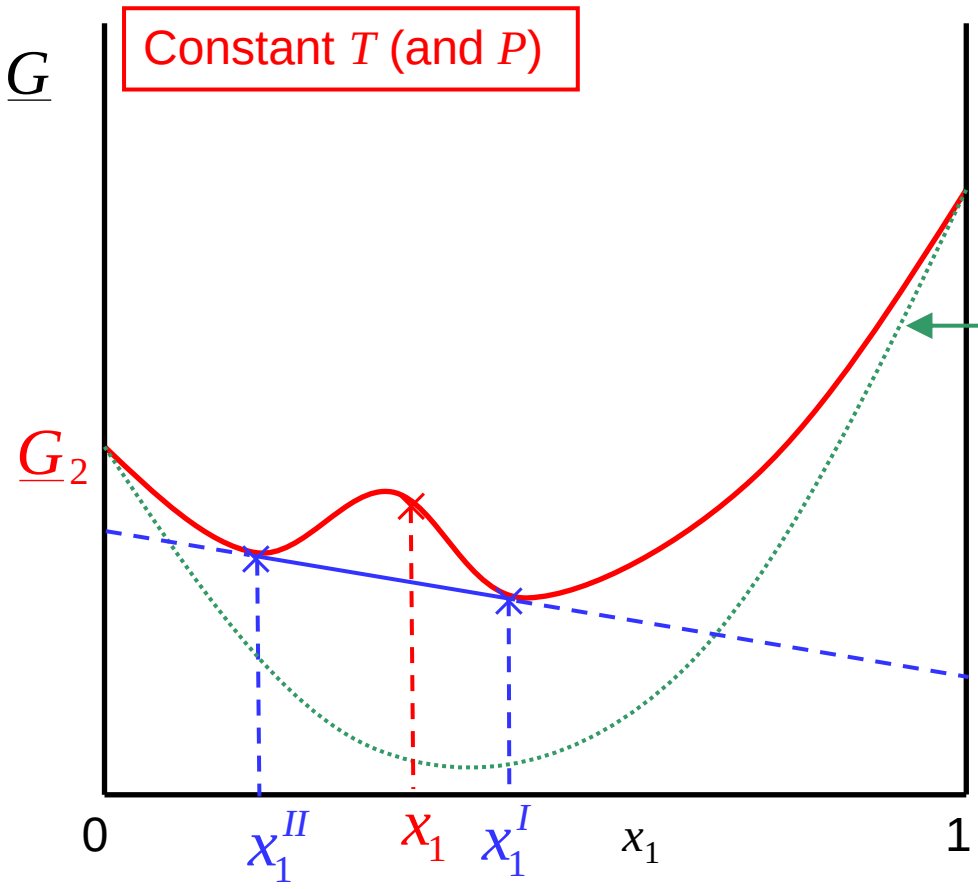
Recap – started to look at Liquid Liquid Equilibrium

Liquid-liquid separation occurs at a particular temperature when two points exist at which $\frac{d^2G}{dx_1^2} = 0$

(equivalent to $\frac{1}{x_1} + \frac{d \ln g_1}{dx_1} = 0 \Rightarrow \frac{dP}{dx_1} = 0$)

Gibbs energy is then minimised by “unmixing” into two separate liquid phases both of which are themselves mixtures, but with different compositions.

REPEAT



$$\underline{G}(x_1) > \lambda^I \underline{G}(x_1^I) + (1-\lambda^I)\underline{G}(x_1^{II})$$

Temperature dependence (L-L phase diagram)

It is commonly observed that the solubility (i.e. miscibility) of partially miscible liquids increases with temperature

Less common, but also possible, are systems for which solubility decreases with temperature

In order to understand these types of behaviour, it is useful to reconsider the various terms in the expression for the Gibbs free energy change of mixing:

$$\Delta \underline{G}^{mix} = \underline{H}^{ex} - T(\Delta \underline{S}_{IM}^{mix} + \underline{S}^{ex})$$

where $\Delta \underline{S}_{IM}^{mix}$ is the entropy of mixing for an ideal mixture:

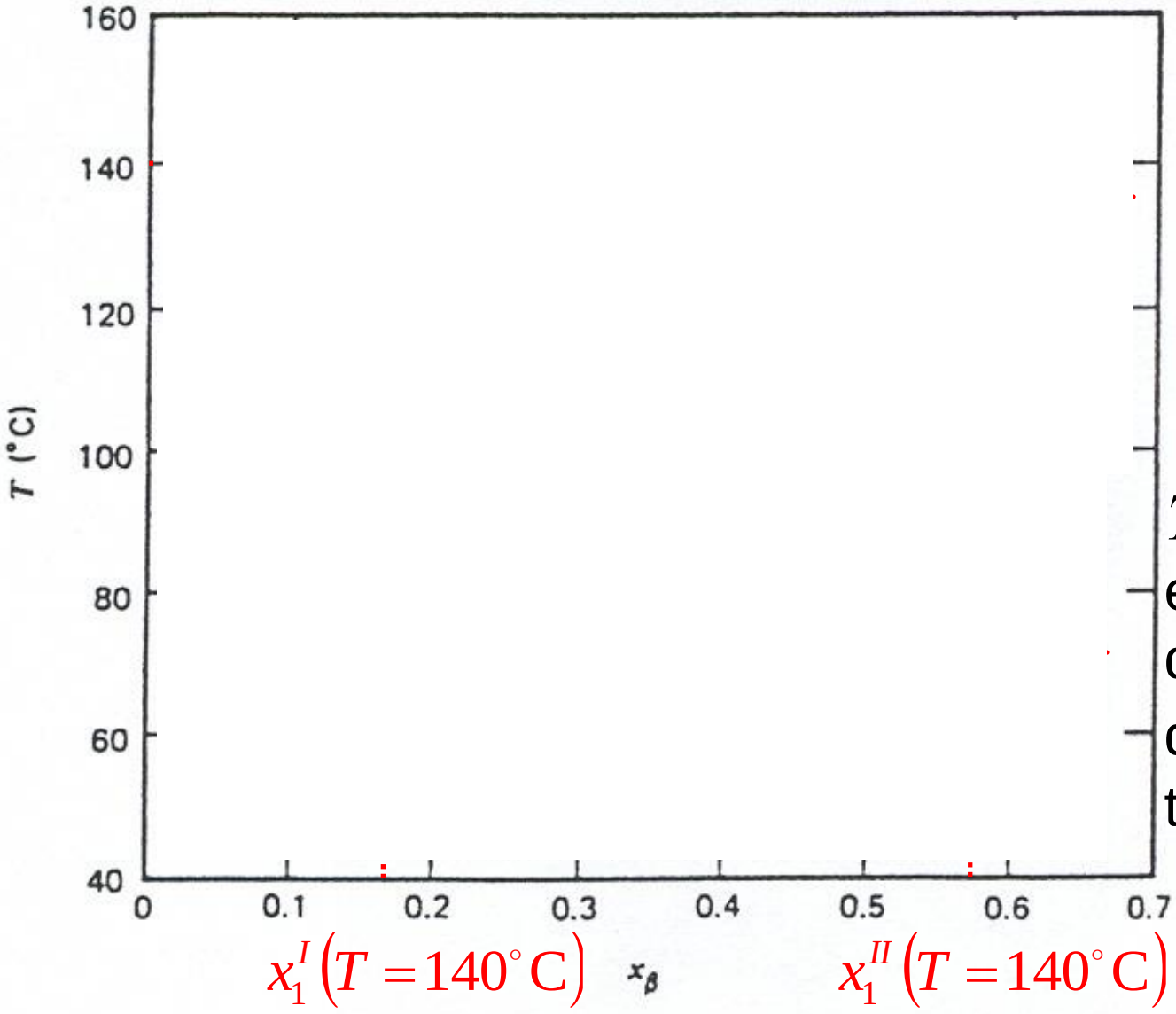
$$\Delta \underline{S}_{IM}^{mix} = -\mathrm{R}\sum x_i \ln(x_i)$$

which is always positive and is independent of temperature

$$\Delta \underline{G}_{IM}^{mix} \propto T$$

We know nothing about how $\underline{G}^{ex}$ will behave with T

(constant P)

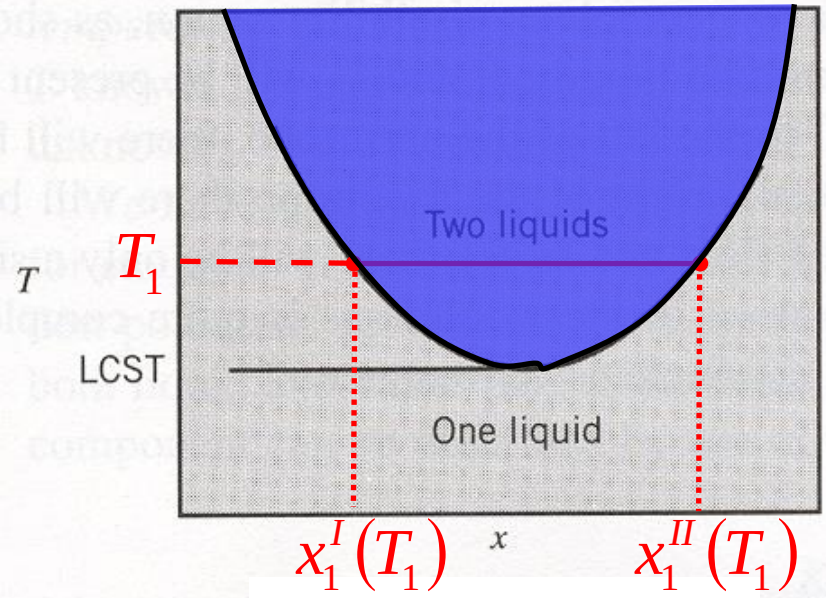


$T\uparrow$ solubility↑
"normal" behaviour
ideal (entropic) contribution dominates

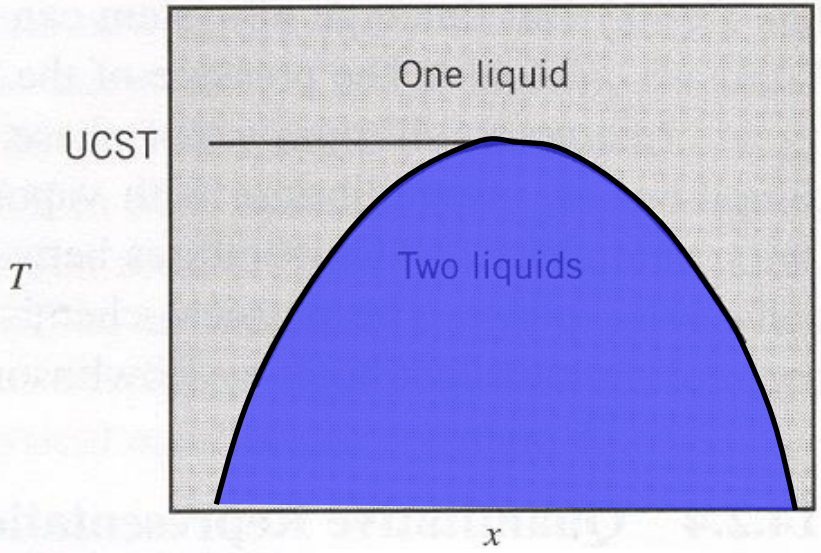
$T\downarrow$ solubility↑
excess term decreases with decreasing T faster than entropic term

[Diagram taken from Sandler, "Chemical and Engineering Thermodynamics", Wiley]

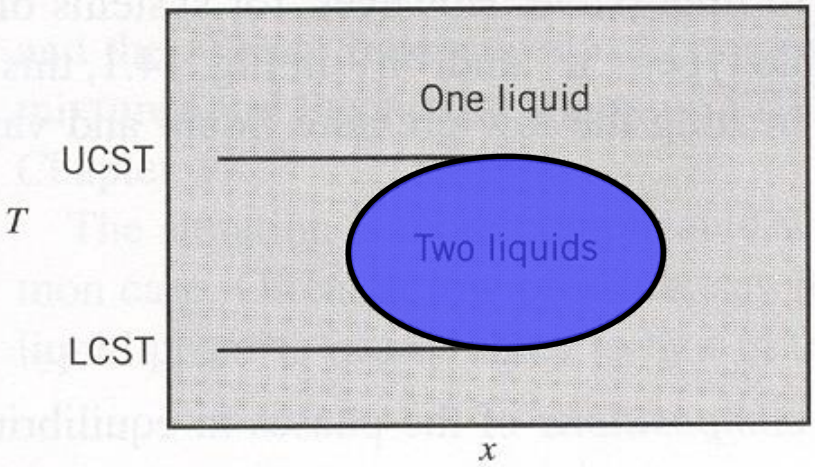
Temperature-composition phase diagram for β-picoline and water



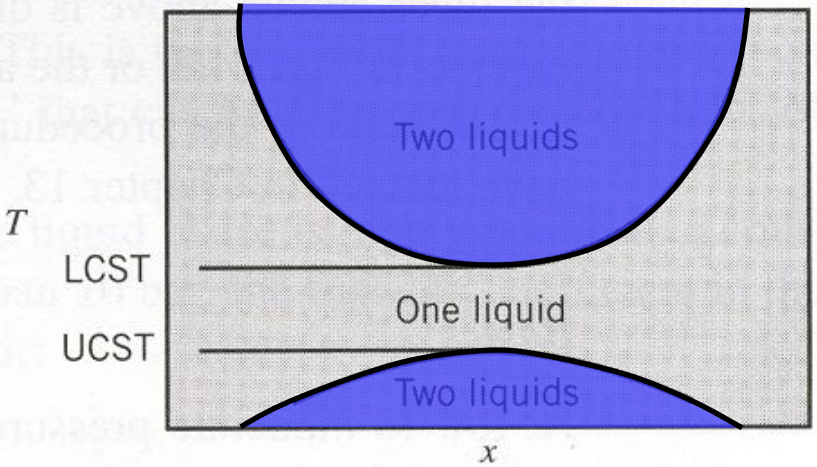
(*a*) triethylamine/water



(*b*) phenol/water



(*c*) deuterium oxide/2-methylpyridine



(*d*) sulphur/triphenylamine

regions of liquid-liquid coexistence

[Diagram taken from Winnick, "Chemical Engineering Thermodynamics", Wiley]

Analogy of single component VLE and binary LLE

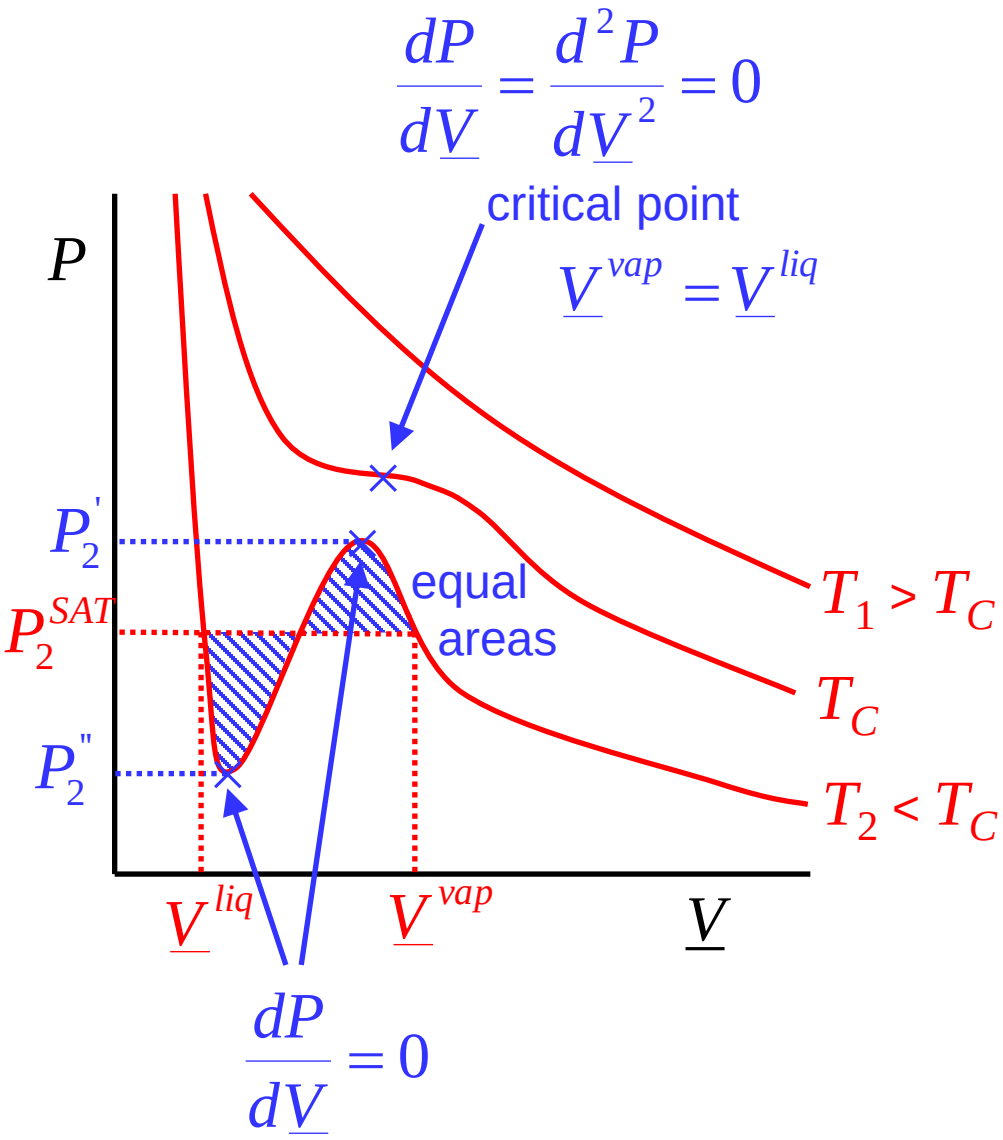
4 DIAGRAMS

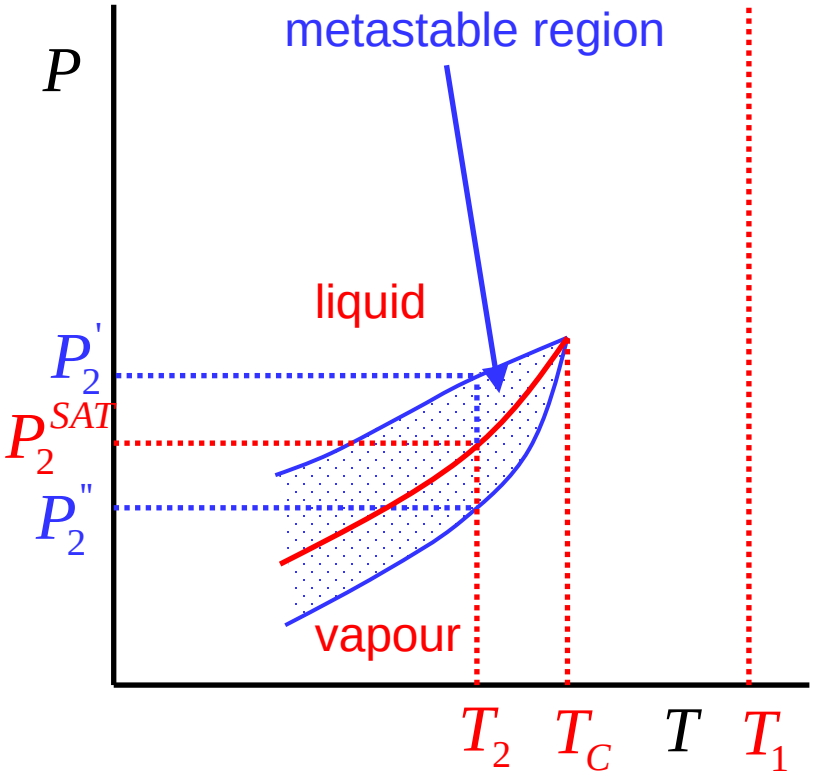
Look at P-$\underline{V}$ isotherms and G-x_1 isotherms as fn(T)

N.b. this is an analogy, not exact equivalence

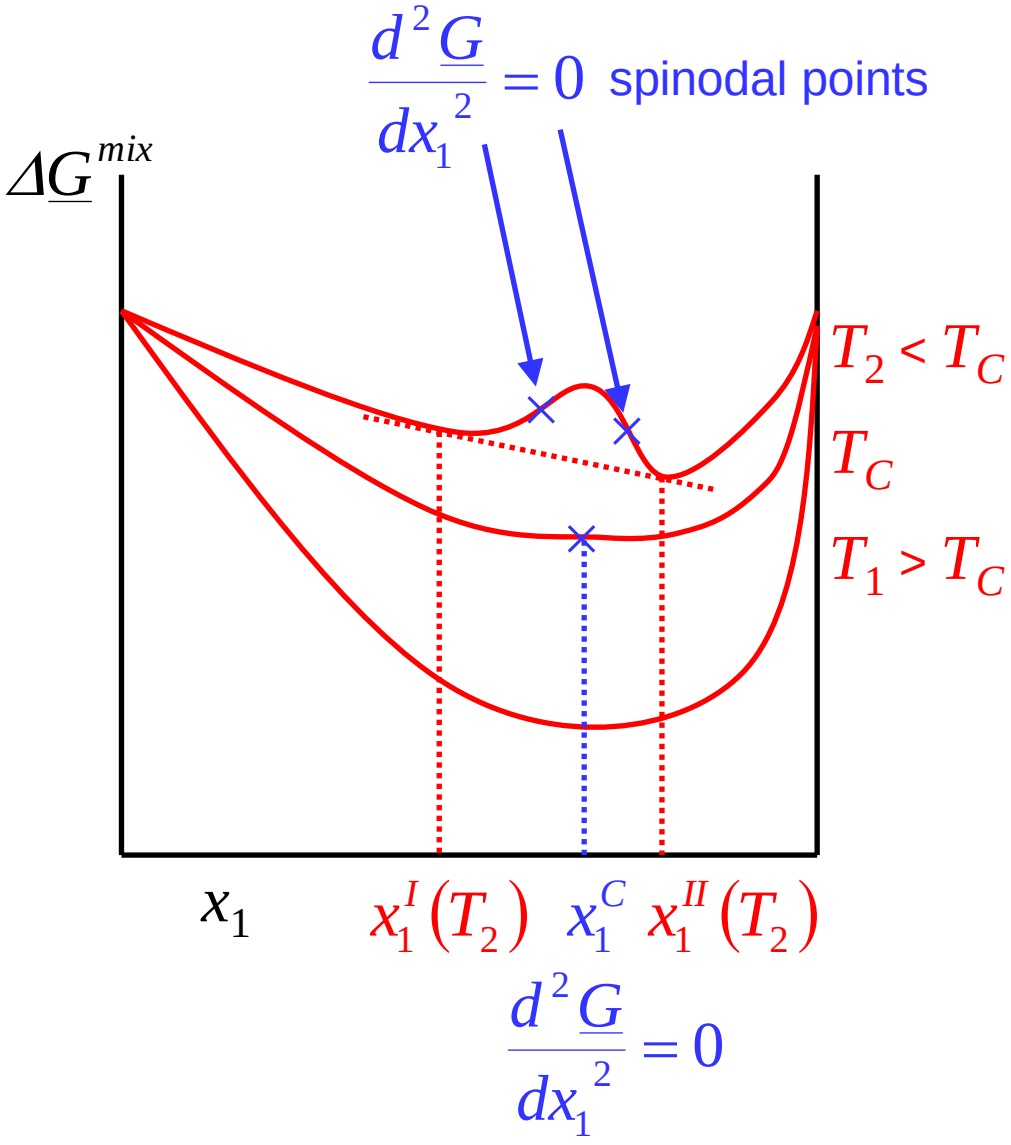
Consolute point $\neq$ Critical point

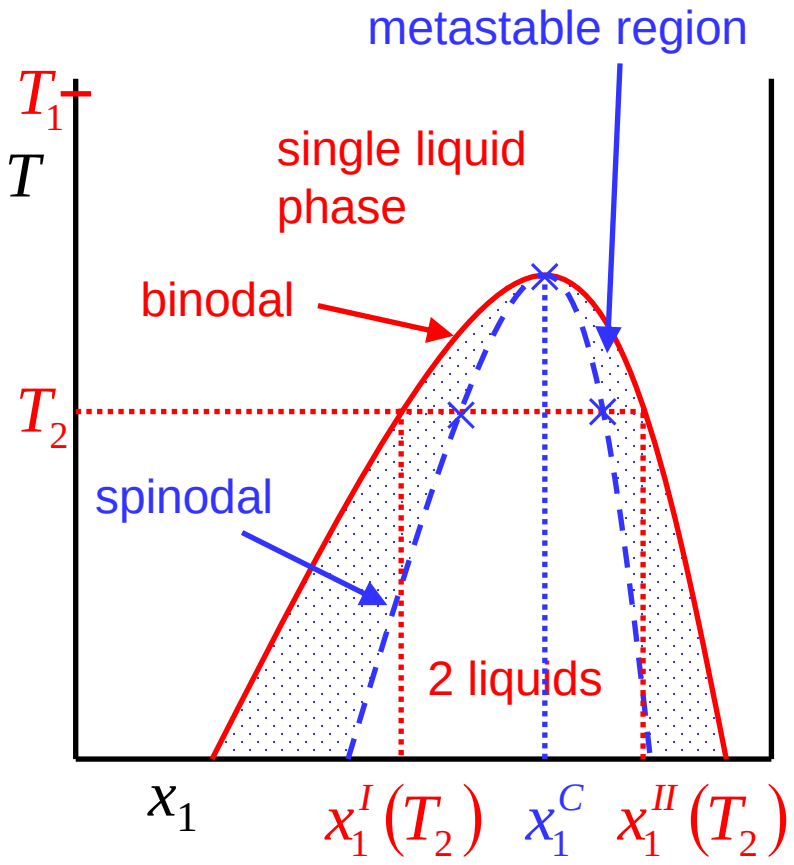
First, consider pure substance VLE:





Now, consider a binary system with an <u>upper</u> consolute point at T_C:





Example

Isobutane (1) and furfural (2) form two liquid phases at 37.8°C and 5 bar

Use the van Laar equations to calculate the compositions of the coexisting phases

Assume $\alpha = 2.62$ and $\beta = 3.02$ are valid at these conditions

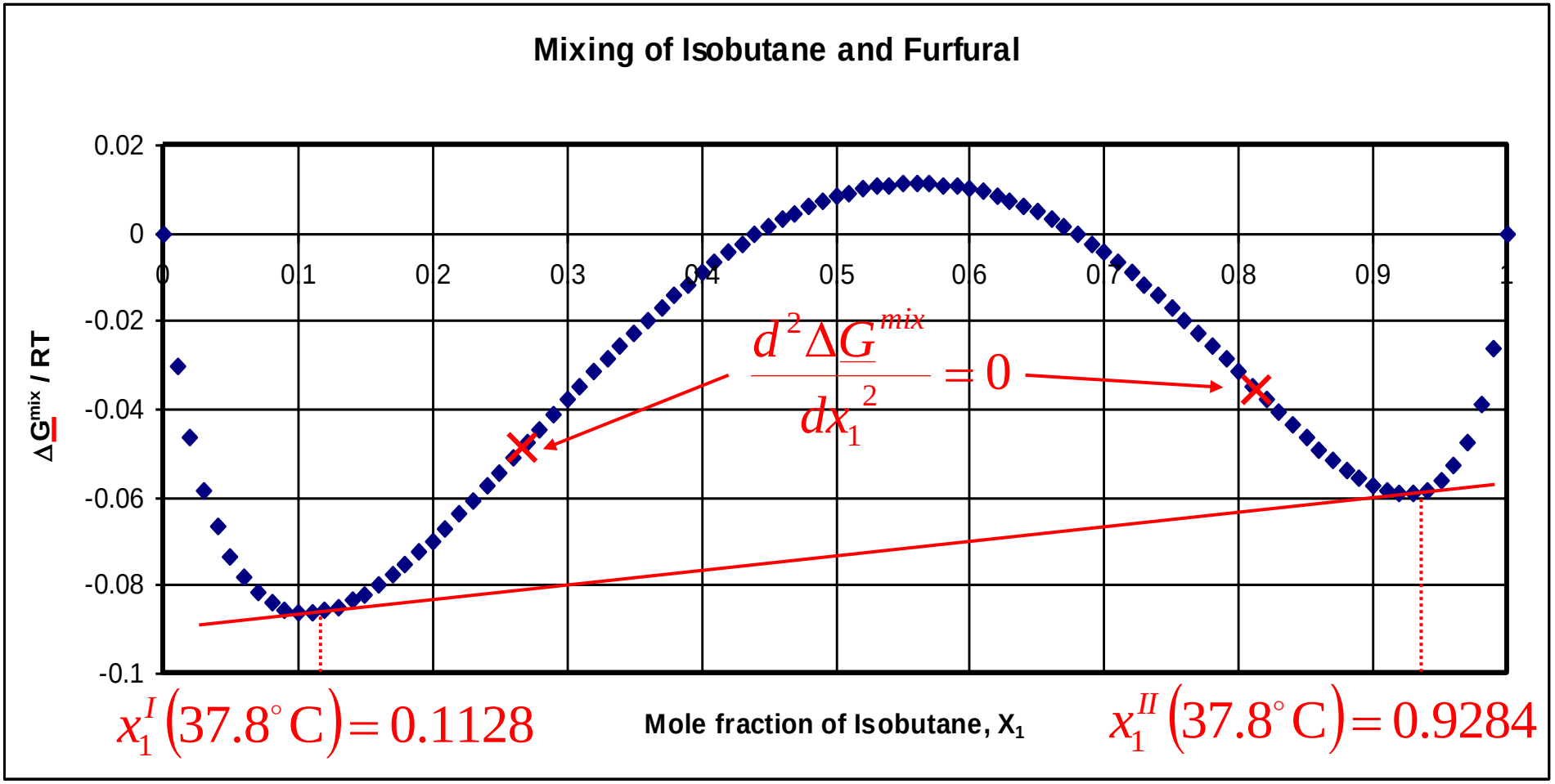
(Note that this example requires a numerical or graphical solution)

N.b. α, β are fns(T)

Use Van Laar equations and put into a spreadsheet:

$$\ln \gamma_1 = \left(\frac{\alpha}{1 + \frac{\alpha x_1}{\beta (1 - x_1)}} \right)^2 \qquad \ln \gamma_2 = \left(\frac{\beta}{1 + \frac{\beta (1 - x_1)}{\alpha x_1}} \right)^2$$

using $\Delta \underline{G}^{mix} = x_1 RT \ln x_1 + x_2 RT \ln x_2 + x_1 RT \ln \gamma_1 + x_2 RT \ln \gamma_2$



N.b. intersections are **not** at minima

The results are:

$$x_1^I = 0.1128 \qquad x_1^{II} = 0.9284$$
$$x_2^I = 0.8872 \qquad x_2^{II} = 0.0716$$
$$\gamma_1^I = 8.375 \qquad \gamma_1^{II} = 1.018$$
$$\gamma_2^I = 1.030 \qquad \gamma_2^{II} = 12.77$$

Thus, for any mixture with 0.1128 < x_1 < 0.9284 two separate phases will form

$$\gamma_1 \to 1 \text{ as } x_1 \to 1$$

The dilute species have very high activity coefficients

At equilibrium $x_1^I \gamma_1^I = x_1^{II} \gamma_1^{II} \Rightarrow \gamma_1^I \approx \dfrac{x_1^{II}}{x_1^I}$ for component 1, dilute in phase I